

LAB - IV
Section: F
Part - II
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Extra Questions

1. Demux

```
-2DemuxFolder > 1-2demux.v
1  module andgate(y,a,b);
2      input a,b;
3      output y;
4      assign y = a & b;
5  endmodule
6
7  module notgate (y,a);
8      input a;
9      output y;
10     assign y = !a;
11 endmodule
12
13 module demux (input wire sel, in, output wire y0,y1);
14     wire n1;
15     andgate a(y1, sel, in);
16     notgate n(n1, sel);
17     andgate b(y0, n1, in);
18 endmodule
```

```

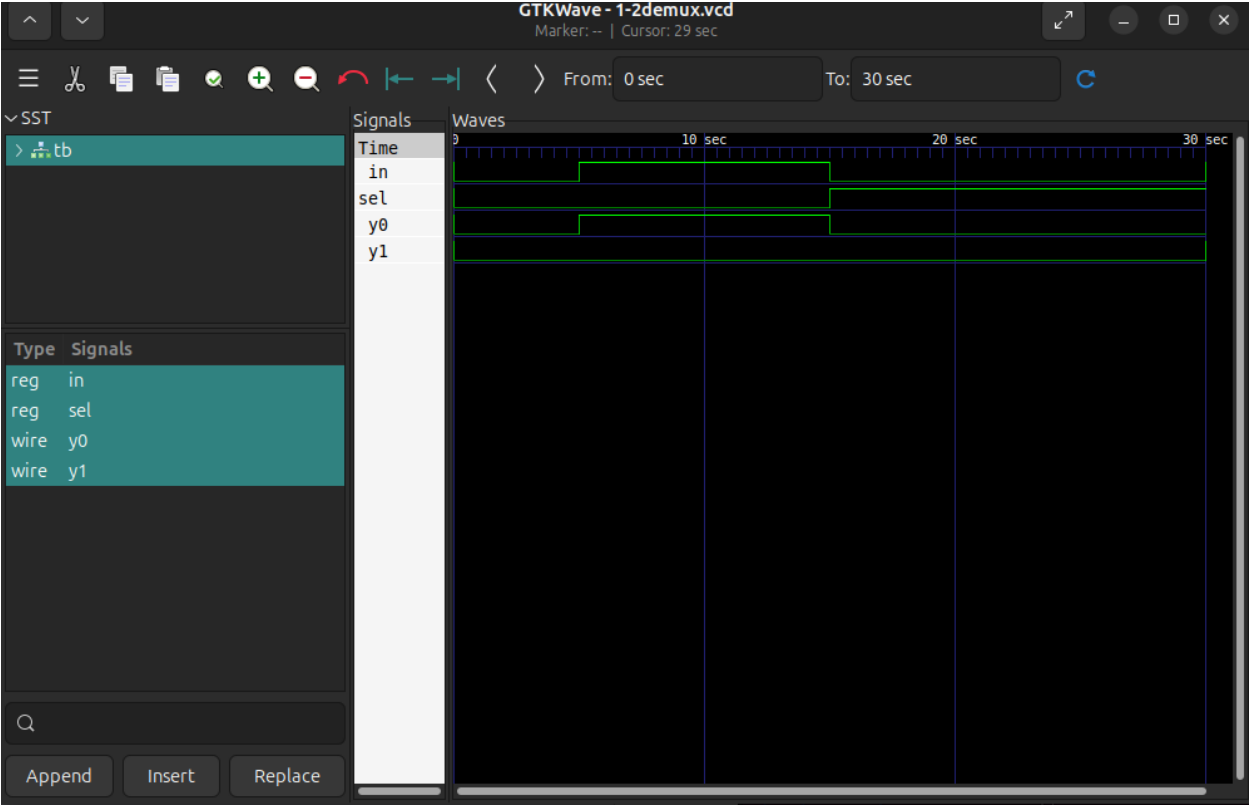
module tb;
    reg sel,in;
    wire y0,y1;
    demux uut(.sel(sel), .in(in), .y0(y0), .y1(y1));
    initial
    begin
        #0 sel=0;in=0;
        #5 sel=0;in=1;
        #10 sel=1;in=0;
        #15 sel=1;in=1;
    end
    initial
    begin
        $monitor($time, "sel=%b, in=%b, y0=%b, y1=%b", sel,in,y0,y1);
    end
    initial
    begin
        $dumpfile("1-2demux.vcd");
        $dumpvars(0,tb);
    end
endmodule

```

```

neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-III/1-2DemuxFolder$ iverilog 1-2demux.v 1-2demux_tb.v -o 1-2demux
neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-III/1-2DemuxFolder$ vvp 1-2demux
VCD info: dumpfile 1-2demux.vcd opened for output.
      0sel=0, in=0, y0=0, y1=0
      5sel=0, in=1, y0=1, y1=0
     15sel=1, in=0, y0=0, y1=0
     30sel=1, in=1, y0=0, y1=1
neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CSE/DD
CO/LAB/LAB-III/1-2DemuxFolder$ █

```



2.Encoder

```
module encoder (input wire [3:0] y, output wire [1:0] a);  
    assign a[1] = y[3] + y[2]; // * MSB  
    assign a[0] = y[3] + y[1]; // * LSB  
endmodule
```

```
module encoder_tb;  
    reg [3:0] Y;  
    wire [1:0] A;  
    initial  
    begin  
        $dumpfile("4-2encoder.vcd");  
        $dumpvars(0, encoder_tb);  
        $monitor("T = %0t Y = %b A = %b", $time, Y, A);  
    end  
    encoder newen(.a(A), .y(Y));  
    initial  
    begin  
        Y = 4'b0001; #5;  
        Y = 4'b0010; #5;  
        Y = 4'b0100; #5;  
        Y = 4'b1000; #5;  
    end  
endmodule
```

```
neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CO/LAB/LAB-III/4-2EncoderFolder$ iverilog 4-2encoder.v 4-2encoder_tb.v -o 4-2encoder
neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CO/LAB/LAB-III/4-2EncoderFolder$ vvp 4-2encoder
VCD info: dumpfile 4-2encoder.vcd opened for output.
T = 0 Y = 0001 A = 00
T = 5 Y = 0010 A = 01
T = 10 Y = 0100 A = 10
T = 15 Y = 1000 A = 11
neeraj-r-rugi@neeraj-ubuntu:/media/neeraj-r-rugi/NEERAJ_DRIVE/PES/3rd-Sem-CO/LAB/LAB-III/4-2EncoderFolder$
```

